Case Study 3: Simple Electricity Bill Calculator

1. Problem Statement

Calculate a household electricity bill based on units consumed. Charge ■2 per unit for the first 100 units and ■3 per unit for units above 100.

2. Objectives

1. Read consumer name and units consumed.
2. Calculate bill using two consumption slabs.
3. Use conditional statements.
4. Display the electricity bill.

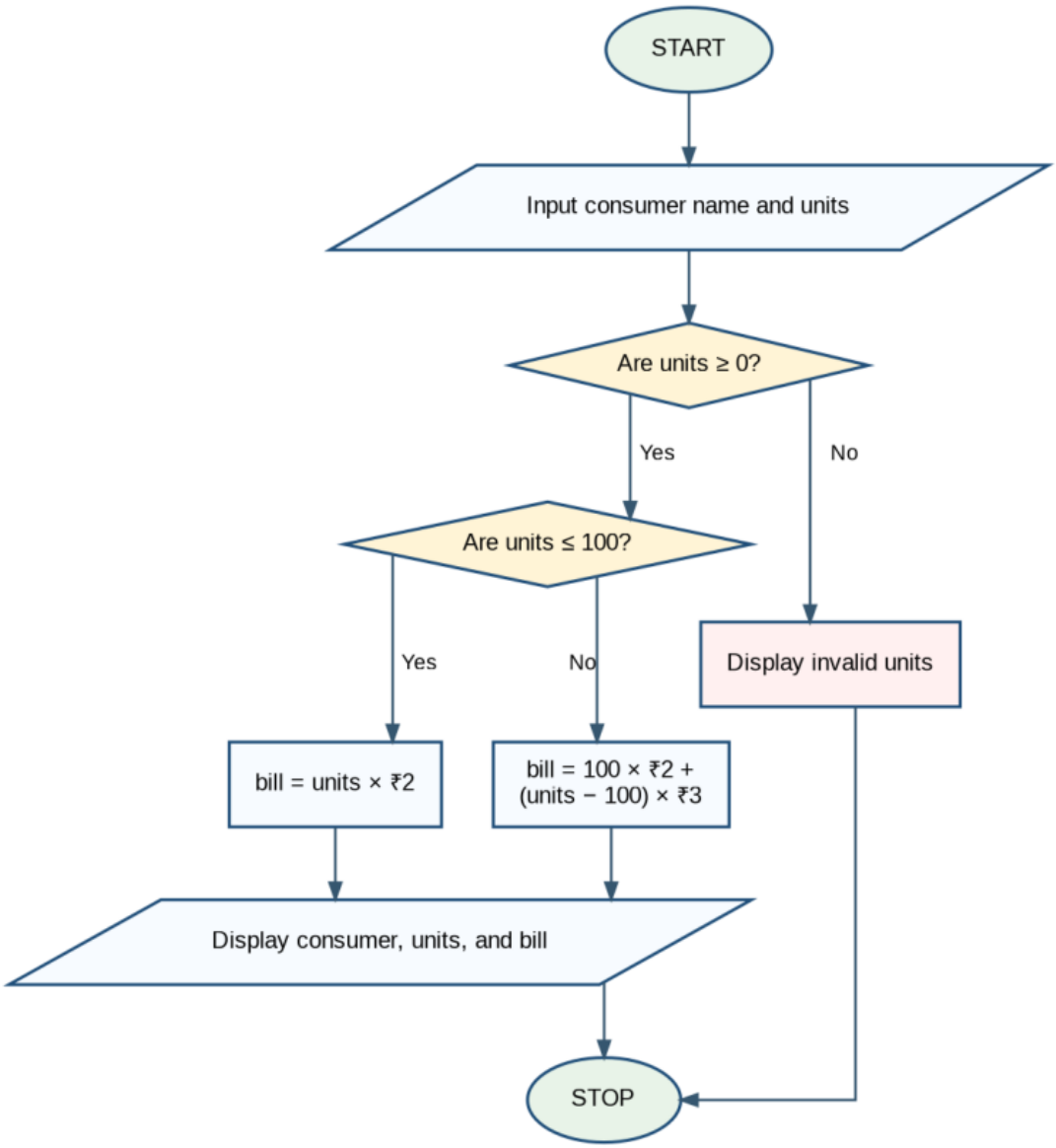
3. Assumptions

1. Units consumed cannot be negative.
2. First 100 units cost ■2 per unit.
3. Units above 100 cost ■3 per unit.
4. No fixed charge or tax is included.

4. Algorithm

1. Start.
2. Read consumer name and units consumed.
3. If units are up to 100, calculate bill as units × 2.
4. Otherwise calculate bill as 100 × 2 plus remaining units × 3.
5. Display consumer name, units, and bill amount.
6. Stop.

5. Flowchart



6. Pseudocode

```
INPUT name, units
IF units <= 100 THEN
    bill = units * 2
ELSE
    bill = 100 * 2 + (units - 100) * 3
DISPLAY name, units, bill
```

7. Complete Python Source Code

```
name = input("Enter consumer name: ")
units = float(input("Enter units consumed: "))
if units < 0:
    print("Invalid units")
elif units <= 100:
    bill = units * 2
else:
    bill = 100 * 2 + (units - 100) * 3
if units >= 0:
```

```
print("\n----- ELECTRICITY BILL -----")
print("Consumer:", name)
print("Units consumed:", units)
print(f"Bill amount: Rs. {bill:.2f}")
```

8. Test Case and Expected Output

Consumer = Ravi; Units = 150.

Bill = (100 × 2) + (50 × 3) = ■350.

Expected bill amount = ■350.

9. Conclusion

The program calculates an electricity bill using slab rates and a conditional statement.